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Design and Optimization of a Rare-Earth-Free PM-Assisted Synchronous Reluctance Motor using Tetrataenite for Light Electric Vehicles

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The rapid electrification of the automotive sector is currently constrained by the geopolitical volatility and high cost of rare-earth materials (REMs), specifically Neodymium and Dysprosium, required for Permanent Magnet Synchronous Motors (PMSMs). While Synchronous Reluctance Motors (SynRMs) offer a robust, REM-free alternative, they traditionally suffer from lower power factor and torque density. This paper investigates a novel topological design for a Permanent Magnet Assisted Synchronous Reluctance Motor (PMa-SynRM) utilizing Tetrataenite (L10-FeNi) an earth abundant, meteorite-derived material with high theoretical energy product as a sustainable alternative to rare-earth magnets. Targeting the Light Electric Vehicle (LEV) sector, this study employs Finite Element Analysis (FEA) to optimize rotor flux- barrier geometry. The research focuses on maximizing the saliency ratio while strategically integrating Tetrataenite elements to enhance air-gap flux density and mitigate torque ripple. The proposed design is benchmarked against a conventional NdFeB-based PMSM and a standard Induction Motor. Preliminary research indicate that Tetrataenite-assisted topologies can achieve competitive torque performance and efficiency, approaching that of rare-earth variants while significantly reducing raw material costs. This research highlights the potential of advanced magnetic materials in enabling the next generation of sustainable, cost-effective electric powertrains.

Keywords: Synchronous Reluctance Motor (SynRM), Tetrataenite (L10-FeNi), Rare-Earth-Free, Electric Vehicles, Finite Element Analysis (FEA).

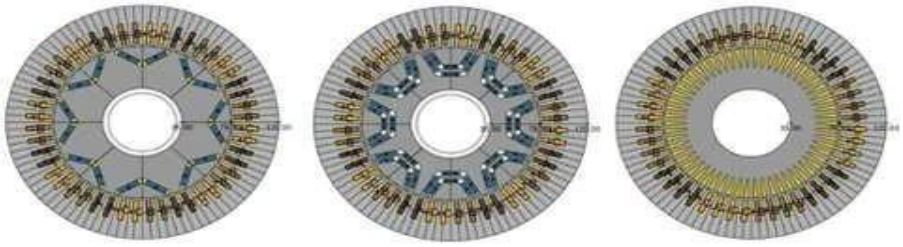


Fig. 1. Geometry examples for PMSM,PMa-SynRM and IM [2]

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